3.1 General properties of waves — 5054 Paper 2

Newest → oldest. Ten remove/add passes on every 5054 Paper 2 from 2015. Only 3.1 subquestions + official MS.

2025

Oct/Nov 5054/22

Question 5 · included (b)(i)

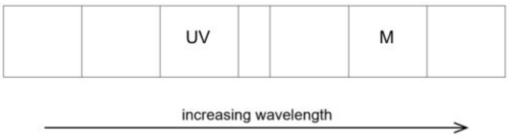
only v=fλ

(i) The speed of microwaves in a vacuum is 3.0×10^8 m/s.

Calculate the wavelength of these microwaves in a vacuum.

wavelength = ... m [3]

Mark scheme — 5054/22 Q5

Question	Answer	Marks
5(a)		B1
5(b)(i)	$(\lambda =)\ \frac{c}{f}$ **or** $\frac{3.0\times10^{8}}{1.2\times10^{N}}$	C1
	2.5×10^{X} (m)	C1
	0.025 (m)	A1
5(b)(ii)	<u>encoded</u> microwave / microwave <u>signal</u> / microwave <u>carrying</u> <u>data</u> / microwave <u>transmitting</u> <u>data</u> (to satellite from Earth)	B1
	satellite transmits microwave to Earth / dish / antenna / subscriber / television	B1
5(c)	to prevent microwaves escaping / leaving the oven / spreading out of the device	B1
	microwaves / they can cause burns (in living tissue / people) / damage to tissue through heating / blisters	B1

Question	Answer	Marks
6(a)	a d.c. / it has one direction **or** does not (repeatedly) reverse direction **or** polarity / direction does not change	B1
6(b)(i)	(Q =) It **or** 41×120	C1
	4.9×10^3 (C)	A1

May/June 5054/21

Question 5 · included (a), (b), (c)

5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.

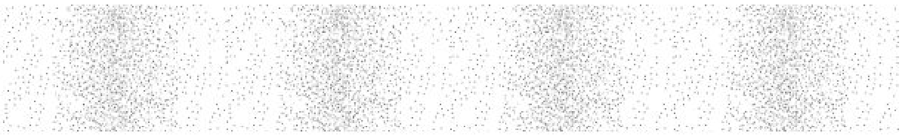


Fig. 5.1

(b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.

(i) The wooden bar makes 45 complete oscillations in 1.0 minute.

Calculate the frequency of the wave produced.

frequency = ... Hz [1]

(ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed ....

wavelength ....
[2]

DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE

(c) Describe what is meant by the diffraction of a water wave.

....

....

.... [1]

[Total: 8]

Mark scheme — 5054/21 Q5

Question	Answer	Marks
5(a)(i)	C and R marked correctly	**B1**
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	**B1**
5(b)(i)	0.75 (Hz)	**B1**
5(b)(ii)	*speed:* no change	**B1**
	wavelength: decreases	**B1**
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other **and** on correct side of normal	**M1**
	refracted towards the surface on correct side of surface **and** join wavefronts shown	**A1**
5(c)	spreading out / bending at an edge / obstacle **or** spreading out after passing through a (small) gap / hole	**B1**

May/June 5054/22

Question 4 · included (a), (b), (c)

(a) Define 'power'.

....

.... [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power = ...W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

DO NOT WRITE IN THIS MARGIN

Mark scheme — 5054/22 Q4

Question	Answer	Marks
4(a)	*motion of particles:* backwards and forwards / to and fro / closer and further apart	B1
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork **or** mention of compressions and rarefactions	B1
4(b)(i)	number of oscillations per second **or** number of wave(length)s (passing a point) per second	B1
4(b)(ii)	2.5 (waves) seen **or** (1 wave takes) 0.02 (s) **or** (f =) 1 / T or (number of) waves / time	C1
	50 (Hz)	A1
4(b)(iii)	quieter / less volume	B1
4(b)(iv)	half frequency shown **and** at least one complete oscillation seen	B1
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	B1
	same wavelength as on left **and** some curving in correct direction	B1
4(c)(ii)	diffraction	B1

2024

Oct/Nov 5054/21

Question 5 · included (b), (c)

not vacuum (sound chapter)

DO NOT WRITE IN THIS MARGIN

(b) Describe how a longitudinal wave differs from a transverse wave.

....

....

....

.... [2]

(c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

Mark scheme — 5054/21 Q5

Question	Answer	Marks
4(c)(ii)	frequency increases **and** wavelength decreases	B1
4(d)	infrared (radiation)	B1

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by vibrating matter / particles	B1
	no matter / particles in a vacuum / no medium	B1
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	B1
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	B1
5(c)(i)	(λ =) v / f **or** 1500 / 8.4 × 10^6	C1
	1.8 × 10^N	C1
	1.8 × 10^{-4} (m)	A1
5(c)(ii)	frequency: does not change	B1
	wavelength: increases	B1

May/June 5054/21

Question 4 · included (a)(i), (a)(ii), (b), (c)

not ultrasound use

(i) State what is meant by 'frequency'.

....

.... [1]

(ii) Calculate the speed of ultrasound in air.

Show your working.

speed = ... m/s [2]

(b) Sound waves are longitudinal waves. Water waves are transverse waves.

(i) Describe, by referring to the movement of the particles in the wave, the difference between a longitudinal wave and a transverse wave.

You may include a labelled diagram to help your description.

....

....

....

....

.... [3]

(c) Fig. 4.1 shows radio waves of long and short wavelength passing over the same hill.

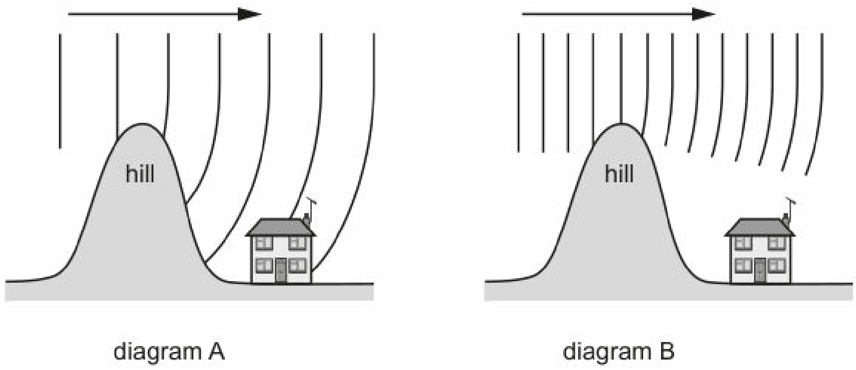


Fig. 4.1 (not to scale)

Explain why the radio waves in diagram A reach the house but the radio waves in diagram B do not reach the house.

....

....

....

.... [2]

[Total: 10]

Mark scheme — 5054/21 Q4

5054/21 Cambridge O Level – Mark Scheme May/June 2024
PUBLISHED

Question	Answer	Marks
3(b)(v)	(liquid) water has a higher specific heat capacity; water has stronger bonds	B1

Question	Answer	Marks
4(a)(i)	number of (complete) waves (passing a point) per second	B1
4(a)(ii)	(V =) $f\lambda$ in any form	C1
	330 m / s	A1
4(a)(iii)	ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance	B1
4(b)(i)	longitudinal wave - oscillation backwards and forwards / in direction of travel of wave	B1
	transverse wave - oscillation at right angles to (direction of travel of) wave	B1
	diagram of each type	B1
4(b)(ii)	earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning	B1
4(c)	long wavelengths bend around hill **and** short wavelengths do not bend	B1
	(due to) diffraction	B1

Question	Answer	Marks
5(a)	correct symbol; 	B1
5(b)(i)	correct curvature	B1
5(b)(ii)	(at higher current / voltage) temperature increases	B1
	(at higher current / voltage) resistance increases	B1

 Page 7 of 10

2023

Oct/Nov 5054/21

Question 4 · included (a), (b)(i)

not n or rays

(a) Calculate the wavelength in air of this red light.

wavelength = ... m [3]

(i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed ....

frequency ....

wavelength ....
[2]

Mark scheme — 5054/21 Q4

Question	Answer	Marks
4(a)	(λ =) v / f **or** 3.0×10^8 / 4.7×10^{14}	C1
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	i = 45° **and** r = 30°	C1
	(n =) sin(i) / sin(r) **or** sin(45°) / sin(30°)	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block **and** refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

Question	Answer	Marks
5(a)(i)	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation)	B1
	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation) **and** in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

Question 5 · included (b)

only trans vs long

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

....

....

.... [2]

[Total: 6]

Mark scheme — 5054/21 Q5

Question	Answer	Marks
4(a)	(λ =) v / f **or** $3.0 \times 10^8 / 4.7 \times 10^{14}$	**C1**
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	**C1**
	6.4×10^{-7} (m)	**A1**
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	**B2**
4(b)(ii)	i = 45° **and** r = 30°	**C1**
	(n =) sin(i) / sin(r) **or** sin(45°) / sin(30°)	**C1**
	1.4	**A1**
4(b)(iii)	straight continuation of ray in block **and** refraction away from normal in air	**B1**
	emergent ray parallel to initial incident ray	**B1**

Question	Answer	Marks
5(a)(i)	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation)	**B1**
	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation) **and** in this order	**B1**
5(a)(ii)	damaging effect: (skin) cancer / cataracts	**B1**
	property: it / ultraviolet radiation is ionising	**B1**

Question	Answer	Marks
5(b)	in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the vibration direction is parallel to the propagation direction	B1

Question	Answer	Marks
6(a)	(because) the current at every point in a series circuit is the same	B1
6(b)(i)	(R =) V / I **or** 1.5 / 0.20 **or** 7.5 (Ω)	C1
	(R_T =) 7.5 – 1.5 –3.5	C1
	2.5 (Ω)	A1
6(b)(ii)	resistance (of T / thermistor) decreases (as the temperature increases)	B1
	current in circuit increases **or** smaller fraction of total resistance	B1
	p.d. across R and S increases **or** (p.d. is) smaller fraction of e.m.f.	B1

Question	Answer	Marks
7(a)(i)	P: (carbon) brush **and** Q: slip ring(s)	B1
7(a)(ii)	to connect the output terminal(s) and the (rotating) coil/brushes	B1
	without wires twisting	B1
7(b)(i)	at least one of the five positions when the output e.m.f. is zero indicated with X **and** no indications elsewhere	B1

Oct/Nov 5054/22

Question 10 · included (b)

only v=fλ

(b) The gamma radiation produced has a frequency of 8.8×10^{19} Hz.

The speed of electromagnetic radiation in a vacuum is 3.0×10^{8} m/s.

Calculate the wavelength of this gamma radiation in a vacuum.

wavelength = ... m [2]

[Total: 9]

Mark scheme — 5054/22 Q10

Question	Answer	Marks
9(b)(iv)	(nuclear reaction produces) a high temperature (in core)	B1
	outward force (due to high temperature / nuclear reaction) **or** force due to high temperature / nuclear reaction **or** radiation pressure	B1
	balances (gravitational) force (inwards)	B1
9(c)	(red supergiant explodes as) supernova (explosion)	B1

Question	Answer	Marks
10(a)(i)	horizontal path marked with γ **and** no other path indicated as γ	B1
10(a)(ii)	beta (β-radiation)	B1
	left-hand rule (mentioned)	B1
	first finger into page **and** thumb downwards	B1
	current opposite to motion **or** particles negative (and so are beta)	B1
10(a)(iii)	beta (β-radiation) completely absorbed	B1
	some gamma rays absorbed **or** (some) gamma passes through	B1
10(b)	(λ =) $v / f = 3.0 \times 10^8 / 8.8 \times 10^{19}$	C1
	3.4×10^{-12} (m)	A1

May/June 5054/21

Question 7 · included (a), (b), (c)

(a) Describe the motion of the glider in the first 12 s.

....

....

.... [2]

(b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = ... kg [3]

(c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = ... m [2]

Mark scheme — 5054/21 Q7

Question	Answer	Marks
7(a)	wavelength has altered / too large **or** crests in shadow should be bent / quarter circular above or below gap **or** vertical lines too straight on two crests	**B1**
7(b)	at least 3 semicircles centered on the gap	**B1**
	wavelength constant and the same on right and left of gap **and** some bending	**B1**
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	**B1**
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	**C1**
	3(.0) Hz)	**A1**
7(c)(iii)	1.3 (cm)	**B1**

May/June 5054/22

Question 5 · included (a), (b)

(a) Underline **two** other examples of transverse waves.

seismic P-waves **seismic S-waves** **sound** **X-rays** [1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

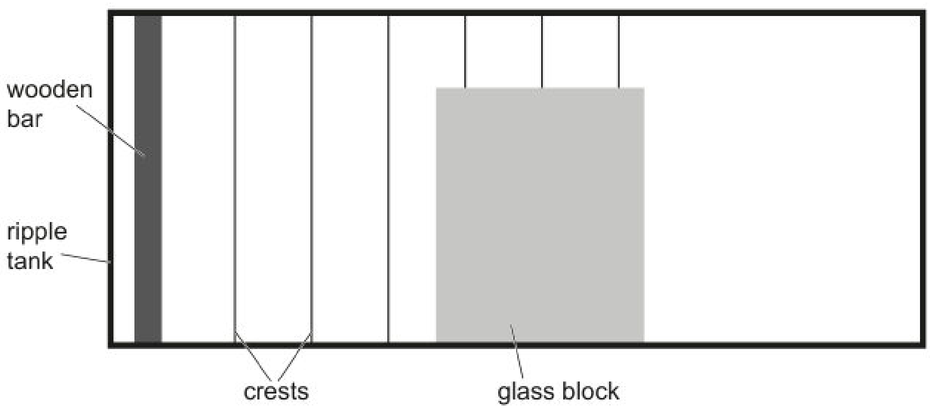


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

Mark scheme — 5054/22 Q5

Question	Answer	Marks
4(a)(i)	–273 (°C) **and** 0 (K)	**B1**
4(a)(ii)	particles) move fast(er)	**B1**
	vibrate / oscillate with larger amplitude	**B1**
4(a)(iii)	internal energy increases	**B1**
	temperature stays the same	**B1**
4(b)(i)	(E =) mcT in any form numerical or algebraic	**C1**
	8000 (J)	**A1**
4(b)(ii)	100–44 **or** 56	**C1**
	0.48 **or** 0.47 (J / (g °C))	**A1**

Question	Answer	Marks
5(a)	seismic S-waves **and** X-rays	**B1**
5(b)(i)	(wavelength =) v / f in any form **or** (distance =) vt **or** f = 1 / T or 1 / 0.15	**C1**
	frequency 6.7 (Hz)	**A1**
	wavelength 4.0 (cm)	**A1**
5(b)(ii)	straight wavefronts above block to the right with same wavelength	**B1**
	correct diffraction – wavefronts quarter circles with centre at top corner	**B1**
5(b)(iii)	increase frequency of movement of (wooden) bar	**B1**

Question	Answer	Marks
5(b)(iv)	less diffraction **or** less bending	**B1**

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A)	**B1**
	voltmeter V_2 9(.0 V)	**B1**
6(a)(ii)	(R =) V / I in any form	**C1**
	36 (Ω)	**A1**
6(a)(iii)	current is (directly) proportional to voltage / potential difference	**B1**
	provided the temperature is constant	**B1**
6(b)	0.35 (A) seen **or** total resistance 36 + 6 / 0.35; 36 + 17(.1); 53(.1) Ω	**B1**
	voltage (across R) 0.35 × 36 **or** 12.6 (V) seen	**B1**
	19 V	**B1**

Question	Answer	Marks
7(a)(i)	X-rays **and** gamma (radiation / rays)	**B1**
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	**B1**
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	**B1**

2022

Oct/Nov 5054/21

Question 4 · included (b)(i)

only frequency

(b) Calculate the speed of the car when it reaches B.

speed = ... [3]

Mark scheme — 5054/21 Q4

Question	Answer	Marks
4(a)	electromagnetic (waves) **or** components of electromagnetic spectrum	B1
4(b)(i)	number of oscillations / cycles / wavelengths	B1
	per unit time	B1
4(b)(ii)	gamma rays	B1
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	B1
	X-rays pass through flesh but not (to same extent) through bones	B1
	image produced by digital / electronic / CCD / CMOS detector	B1
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage **or** X-rays show the bone structure rather than foetal development **or** image would not be clear	B1

Oct/Nov 5054/22

Question 4 · included (b), (c)

not vacuum exp; not hearing

(b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength g is 10 N / kg.

(c) Describe what is meant by 'a rarefaction'.

....

.... [1]

Mark scheme — 5054/22 Q4

Question	Answer	Marks
3(a)	chemical potential energy **and** no other energy	**B1**
3(b)(i)	radiation **or** infrared mentioned	**B1**
	radiation **and** infrared (travel to the man)	**B1**
	energy / infrared / radiation absorbed by the man / pullover **or** travels in straight line **or** travels at the speed of light **or** increases (man's) internal energy	**B1**
3(b)(ii)	(shiny surface) reflects radiation (to man)	**B1**
	less (thermal) energy escapes **or** more (thermal) energy reaches / absorbed by man	**B1**
3(b)(iii)	**EITHER** black (surface) is good absorber / poor reflector of infrared radiation	**B1**
	more (thermal) energy absorbed **or** more thermal energy transferred to man	**B1**
	OR trapped air **and** reduced convection / air poor conductor / air is a good insulator	**(B1)**
	less (thermal) energy lost	**(B1)**

Question	Answer			Marks
4(a)	equipment	e.g.	(evacuable) container **and** (suspended electric) bell / source of sound **and** (air) pump	**B1**
	action	e.g.	ringing / sound (produced) / sound heard **and** evacuate air / vacuum (produced)	**B1**
	observation	e.g.	transmission / hearing of sound ceases / no sound	**B1**

Question	Answer	Marks
4(b)	(they) vibrate	M1
	(vibration) parallel to / in the direction of propagation / of energy travel	A1
4(c)	(point where) molecules are further apart (than average) **or** (point where) pressure / density is lower (than average)	B1
4(d)	(λ =) v / f **or** 1500 / 140 000	C1
	1.1×10^{-2} m **or** 1.1 cm **or** 11 mm	A1

Question	Answer	Marks
5(a)	energy / work done per unit charge **or** $\frac{\text{energy}}{\text{charge}}$ **or** $\frac{\text{work done}}{\text{charge}}$	B1
	property of a source / cell / battery / power supply **or** energy used driving charge around a (complete) circuit **or** energy transferred to electrical energy	B1
5(b)	it lasts for a longer time **or** more (electrical) energy available **or** power supply still functions if one cell fails / runs flat **or** can replace one cell without stopping circuit / stopping current	B1
5(c)(i)	resistance (of thermistor / circuit) decreases **c.a.o.**	B1
	current change **or** voltage change across thermistor matching the change in the resistance of the resistor	B1
	reading of voltmeter increases **c.a.o.**	B1
5(c)(ii)	(V_1 =) $V_0 \times R_1 / R_0$ **or** $1.5 \times 4000 / (4000 + 8000)$ **or** $1.5 \times 4000 / 12\,000$ **or** $I = V / R$ **or** $1.5 \times 8000 / (4000 + 8000)$ **or** I =1.5 / 12 000 **or** 0.125 mA	C1
	0.50 V	A1

May/June 5054/22

Question 8 · included (a), (b), (c)

not satellite

8 (a) Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

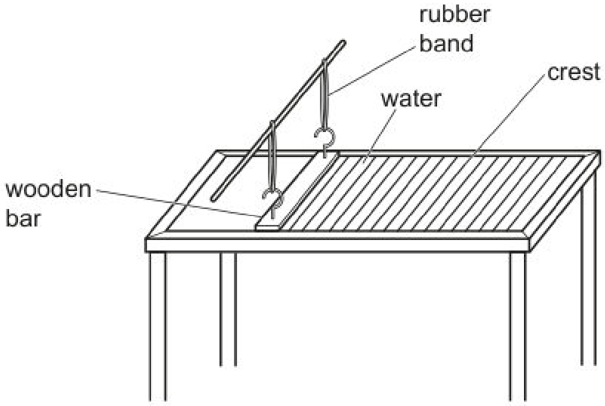


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

(b) **(i)** The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

....

.... [1]

(c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

(i) State the additional apparatus needed to demonstrate refraction.

.... [1]

(ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

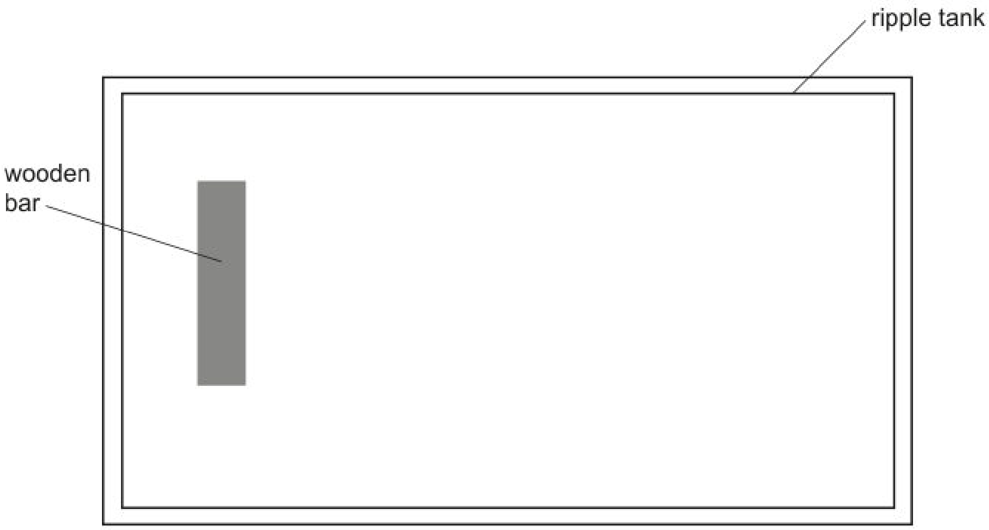


Fig. 8.3

[3]

(b) **(i)** The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

....

.... [1]

(ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed ....

wavelength ....
[2]

(c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

(i) State the additional apparatus needed to demonstrate refraction.

.... [1]

(ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

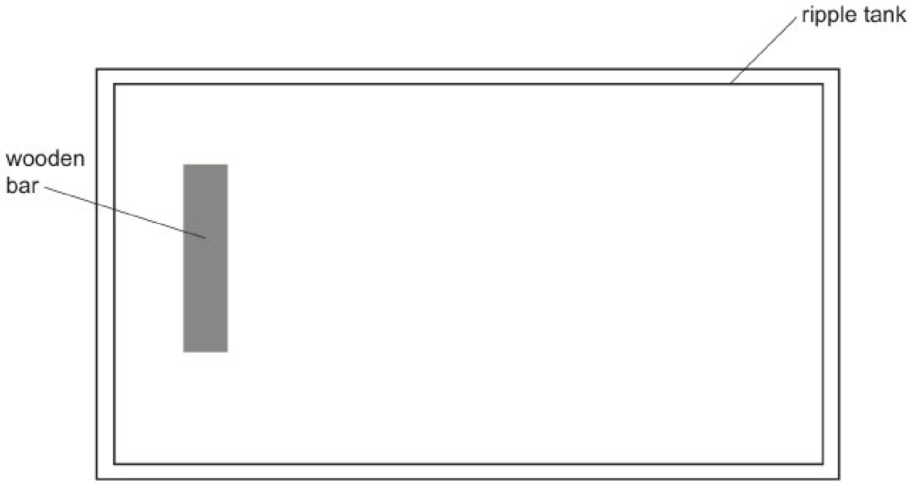


Fig. 8.3

[3]

Mark scheme — 5054/22 Q8

Question	Answer	Marks
8(a)(i)	3	**B1**
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	**B1**
	1 wave taking 0.5 s for first wave shown within half a square **or** 1.5 s shown for the number of waves in **(a)(i)**	**B1**
8(b)(i)	move bar up and down more often (per second) / faster **or** make rubber band(s) strong(er) / short(er) / tight(er) / more in number **or** use bar of less mass / weight	**B1**
8(b)(ii)	*speed* remains the same	**B1**
	wavelength decreases	**B1**
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate **or** a shallow(er) / deep(er) area / boundary	**B1**
8(c)(ii)	diagram showing within the ripple tank: • crests parallel to bar hitting boundary with a boundary labelled	**B1**
	• correct refraction of plane crests either towards or away from normal on the same side of the normal **or** no direction change if crests are initially parallel to boundary but with a wavelength change	**B1**
	• different wavelength shown before and after refraction and constant before and after (by eye)	**B1**
8(d)(i)	(t =) distance / speed algebraic or numerical seen in any form	**C1**
	2 × 560(000) seen or used **or** 0.0019 (s)	**C1**
	0.0037 s	**A1**

Question	Answer	Marks
8(d)(ii)	One of (both): • reflect / refract • have same speed (in a vacuum) • travel <u>in a vacuum</u> / don't need a medium • are electromagnetic • are transverse • carry energy / cause heating • have larger wavelength / smaller frequency than visible light	**B1**
8(d)(iii)	less likely to be hacked / intercepted / greater security **or** more channels / better bandwidth possible / more data **or** less likely for transmission to be interrupted **or** cost of <u>satellites</u> saved / <u>satellites</u> expensive	**B1**

Question	Answer	Marks
9(a)(i)	helium atom has (any 1 from) • two electrons (and ion has 1 electron) • one more electron (than the ion) • no charge (and ion positive) • same number of electrons as protons;	**B1**
9(a)(ii)	alpha particle has (any 1 from) : • no electrons • charge +2 (e) • only protons and neutrons **or** He ion has an electron	**B1**

2021

Oct/Nov 5054/21

Question 8 · included (b)(iii), (b)(v)

only long/trans and v=fλ

(b) An uncharged, conducting sphere is suspended from an insulating thread. The positively charged rod is placed near to the sphere, as shown in Fig. 5.1.

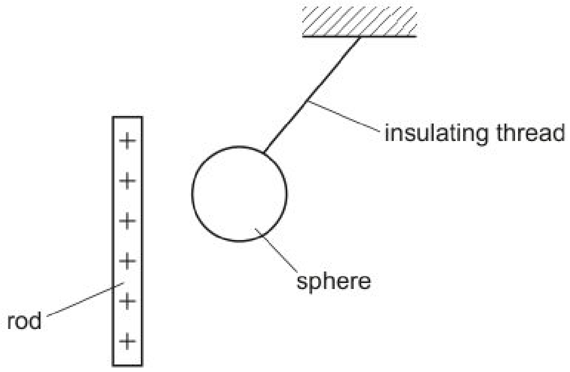


Fig. 5.1

(i) On Fig. 5.1, draw the distribution of charge on the sphere. [2]

(ii) Explain why the sphere is pulled towards the rod.

....

....

....

.... [2]

[Total: 6]

Mark scheme — 5054/21 Q8

Question	Answer	Marks
7E(a)	magnetic field lines cut the solenoid / coils **or** changing magnetic field in solenoid	**B1**
	an electromotive force / e.m.f. is induced	**B1**
	electromotive force / e.m.f. produces the current	**B1**
7E(b)	negative / opposite / reversed current / deflection / reading	**B1**
	negative e.m.f. / voltage (induced) **or** magnetic field cut in opposite direction **or** magnetic field changes in opposite sense / decreases (owtte)	**B1**
7O(a)	NOR (gate)	**B1**
7O(b)(i)	lower logic gate has an input of logic level 1 (and so its output / Y is 0) **or** the output X has a logic level 1 and so the inputs must both be 0 (Y is an input to the upper NOR gate)	**B1**
7O(b)(ii)	X changes to 0 **and** Y changes to 1.	**B1**
7O(b)(iii)	both X and Y stay at the new values **or** neither X nor Y changes	**B1**
7O(b)(iv)	the output shows which terminal (P or Q) last had the value 1.	**B1**

Question	Answer	Marks
8(a)(i)	magnetic field **or** magnet	**B1**
8(b)(i)	the current (in coil continually) reverses (direction) **or** the force (on the coil continually) reverses	**B1**
	the current (in the coil continually) reverses (direction) **and** the force (on the coil continually) reverses	**B1**

Question	Answer	Marks
8(b)(ii)	cone moves forwards **and** air is compressed / hits (air) molecules	**B1**
	cone moves backwards **and** rarefaction produced / molecules sucked back	**B1**
	(air) molecules vibrate **or** production of compressions and rarefactions continues	**B1**
8(b)(iii)	vibration direction is parallel to energy travel / propagation direction	**B1**
	in transverse wave, vibration direction is perpendicular to propagation / energy travel direction	**B1**
8(b)(iv)	(time period =) 0.0050 (s) **or** 1 / 0.0050 **or** 200 (times per second)	**C1**
	400 (times per second)	**A1**
8(b)(v)	(λ =) c / v **or** 340 / 200	**C1**
	1.7 m	**A1**
8(c)(i)	decreases / less loud / quieter	**B1**
	amplitude of sound / vibration decreases **or** (maximum) force on coil decreases	**B1**
8(c)(ii)	(pitch) stays the same / unchanged **and** no change to frequency / time period	**B1**

Question	Answer	Marks
9(a)(i)	molecules with similar / slightly smaller packing density **and** most molecules touching	**B1**
	random arrangement	**B1**
9(a)(ii)	molecules (of gas) further apart	**B1**
	(repulsive) forces between molecules (much) smaller	**B1**

Oct/Nov 5054/22

Question 3 · included (a)

not TIR

(a) Light travels more slowly in the block than in air.

Mark scheme — 5054/22 Q3

5054/22 Cambridge O Level – Mark Scheme October/November 2021
PUBLISHED

Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	**B1**
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	**B1**
	(at constant speed) forward force is equal to resistive / backward force **or** no resultant force **or** forces balance	**B1**

Question	Answer	Marks
3(a)(i)	it / light refracts / bends / moves towards normal **or** angle of incidence greater than angle of refraction	**B1**
	change in direction of the ray / refraction shows a change in speed (at the boundary) **or** sin(*i*) / sin(*r*) is ratio of speeds **or** top of ray slows down first **and** enters the block first	**B1**
3(a)(ii)	(wavelength) decreases	**B1**
	(frequency) does not change.	**B1**
3(b)(i)	(sin(*r*) =) sin(*i*) / *n* **or** sin(45°) / 1.6 **or** 0.71 / 1.6 **or** 0.44 **or** (*r* =) 21(°)	**C1**
	26(°)	**C1**
	64°	**A1**
3(b)(ii)	64° / angle of incidence / *i* is greater than the critical angle / 39° **or** angle of incidence = 64°	**B1**
	it undergoes total internal reflection	**B1**

Question	Answer	Marks
4(a)(i)	13 A	**B1**
4(a)(ii)	0 (A)	**B1**

 Page 6 of 11

May/June 5054/21

Question 4 · included (a)

not table of other topics

4 **(a)** Fig. 4.1 shows a wave on a rope and Fig. 4.2 shows a wave on a spring. Both waves are moving in the direction shown by the arrows.

Fig. 4.1

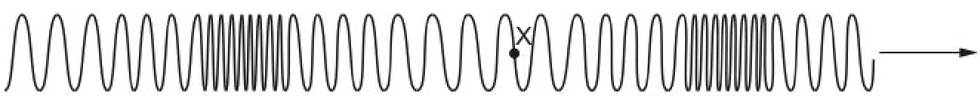


Fig. 4.2

Mark scheme — 5054/21 Q4

Question	Answer	Marks
3(a)	(E =) mcT in any form algebraic or numerical	C1
	2 × 460 × 450	C1
	4.1 ×× 10^5 J	A1
3(b)(i)	any one of: • boiling occurs at one temperature / boiling point **or** evaporation occurs at any temperature / below boiling point • boiling occurs within liquid **or** evaporation occurs on the surface • evaporation increased by draughts, more surface area (and boiling does not)	B1
3(b)(ii)	(average) speed / K.E. of molecules decreases (as temperature falls) **or** (molecules) do not have the energy to escape / break bonds	B1
	fewer fast / energetic molecules escape **or** (only) fastest / high KE molecules escape	B1

Question	Answer	Marks
4(a)(i)	Fig.4.1 transverse **and** Fig.4.2 longitudinal	B1
4(a)(ii)	wave drawn with larger wavelength	B1
4(a)(iii)	backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement)	B1
4(b)	frequency	B1
	ultrasound	B1
	gamma	B1

May/June 5054/22

Question 10 · included (a)

not spectrum

10 (a) Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

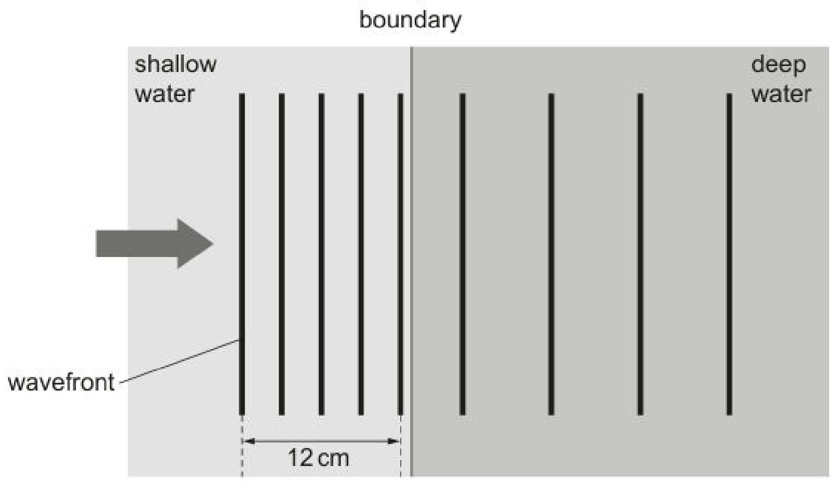


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

Mark scheme — 5054/22 Q10

Question	Answer	Marks
9(c)(i)	(E =) mcT numerical or algebraic in any form	C1
	0.11 × 830 × 5	C1
	460 J	A1
9(c)(ii)	energy output / energy input	C1
	useful energy output / (total) energy input	A1
9(c)(iii)	(energy input =) 460 + 5200 **or** 5660 (J) seen **or** 5200 ÷ (energy in **9(c)(i)** + 5200)	C1
	92% **or** 0.92	A1
9(c)(iv)	(thermal) energy <u>lost</u> / heat <u>lost</u> (from battery to environment)	B1
	energy input larger than calculated / 5660 (J) **or** temperature rise should be larger / is smaller than it should be **or** thermal energy / heat produced larger than 460 J / calculated	B1

Question	Answer	Marks
10(a)(i)	3 (.0) cm	B1
10(a)(ii)	number of waves per second **or** 5 / 10 **or** f = 1 / T	C1
	2(.0) Hz	A1
10(a)(iii)	(v =) $f\lambda$ **or** s = d / t numerical or algebraic in any form	C1
	6(.0) cm / s	A1
10(a)(iv)	frequency stays the same **and** speed increases	B1

Question	Answer	Marks
10(a)(v)	⩾ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	B1
	constant wavelength shown between all refracted wavefronts	B1
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	B1
10(b)(i)	four colours from violet, indigo, blue, green, yellow, orange, red	B1
	correct order of four of the above colours	B1
10(b)(ii)	prism	B1
	slit (to produce narrow beam)	B1
	dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face	B1
	correct refraction on first and second faces for all rays shown	B1

Question	Answer	Marks
11(a)	electron	B1
11(b)	prevents exposure to radiation / for safety (of experimenter and others)	B1
	beta particles stopped / absorbed by metal / case **or** prevents radiation travelling in all directions / to surroundings	B1
11(c)(i)	decreases (slightly) / little change	B1
	particles spread out **or** <u>some</u> stopped / absorbed (by air)	B1
11(c)(ii)	no particles detected **or** <u>large</u> decrease	B1
	(>1m) <u>air</u> stops (beta) particles / <u>air</u> atoms / particles become ionised	B1

2020

Oct/Nov 5054/21

Question 4 · included (a)

not TIR

(a) Light travels more slowly in glass than in air.

Mark scheme — 5054/21 Q4

5054/21 Cambridge O Level – Mark Scheme October/November 2020
PUBLISHED

Question	Answer	Marks
3(a)(i)	(Γ =) $Fx_{\perp r}$ **or** 25 × 0.72	**C1**
	18 N m	**A1**
3(a)(ii)	(WD =) $Fx_{\parallel}$ **or** $F\pi r / 2$ or 25 × 4.5 / 4 **or** 25 × 1.13	**B1**
	28 J	**B1**
3(b)	moment due to force *P* is greater	**B1**
	distance of X to hinge is greater than distance from where force *F* acts	**B1**

Question	Answer	Marks
4(a)(i)	it decreases	**B1**
4(a)(ii)	it does not change	**B1**
4(b)(i)	(c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$ **or** $\sin^{-1}(0.625)$	**C1**
	39°	**A1**
4(b)(ii)	reflected ray at Z **and** $i = r$ **and** no refracted ray	**B1**
	ray strikes vertical face perpendicularly **and** undeviated in air	**B1**

Question	Answer	Marks
5(a)	X, Y and Z to be two (or three) different metals	**B1**
5(b)(i)	(output / thermoelectric) electromotive force / e.m.f. / voltage	**B1**
5(b)(ii)	linearity indicates whether the output (voltage) is directly proportional to something	**B1**
	directly proportional to the temperature difference (between the junctions)	**B1**

 Page 6 of 10

May/June 5054/22

Question 10 · included (b), (c)(i), (c)(ii)

not thermionic/pitch/diode

(b) Fig. 10.1 is the trace on the screen of the oscilloscope.

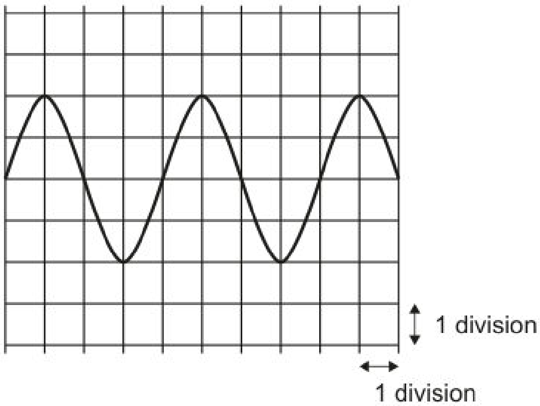


Fig. 10.1

The settings on the oscilloscope are 10 ms / division for the *x*-axis and 3.0 V / division for the *y*-axis.

(c) The trace shown in Fig. 10.1 is caused by a sound.

The sound travels through the air to a microphone from the place that it is made. The microphone is connected to the oscilloscope which displays the waveform shown.

(i) Sound is a type of wave.

State which type.

.... [1]

(ii) Describe the motion of the air molecules as the sound passes through the air to the microphone.

....

....

.... [2]

Mark scheme — 5054/22 Q10

Question	Answer	Marks
9(b)(i)	at least two lines inside coil and horizontal near centre and covering at least six wires of coil	B1
	correct shape of lines fanning out at ends of coil	B1
	at least two lines passing outside the coil from one end to the other	B1
	correct direction of field lines on at least one line and none wrong	B1
9(b)(ii)	N-pole on right hand side	B1
9(c)(i)	move magnet into / near coil **or** move coil	C1
	move magnet / coil quickly **or** move strong magnet	A1
9(c)(ii)	lines of magnetic field cut the coil **or** flux in coil changes	B1
	induced e.m.f. / voltage produced (in complete circuit)	B1

Question	Answer	Marks
10(a)(i)	filament is hot / heated	B1
10(a)(ii)	attracted by a positive potential **or** repelled by filament **or** filament is negative	B1
10(a)(iii)	$2000 \times 1.6 \times 10^{-19}$	C1
	3.2×10^{-16} J	A1
10(b)(i)	two divisions	C1
	6(.0) V	A1
10(b)(ii)	40 ms	B1
10(c)(i)	longitudinal	B1

Question	Answer	Marks
10(c)(ii)	oscillation described, e.g. back and forth	B1
	in direction of travel (of wave / energy) **or** compressions and rarefactions stated or described	B1
10(c)(iii)	more waves seen on screen	B1
	higher frequency (of sound) **or** more waves per second **or** time for one wave less	B1
10(d)(i)	diode passes current in only one direction **or** current only flows for half the cycle / oscillation **or** current is d.c. / direct	B1
10(d)(ii)	at least one top half and no bottom half **or** one bottom half and no top half	C1
	three top halves of the wave and flat sections along mid line **or** two bottom halves of the wave and flat sections along mid line	A1

2019

Oct/Nov 5054/21

Question 10 · included (d)

only wavefront refraction

(iv) Determine the magnification of the image.

magnification = ... [1]

(v) State **two** ways in which a virtual image differs from a real image.

1. ....

....

2. ....

.... [2]

(vi) State one use for a diverging lens.

....

.... [1]

(d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

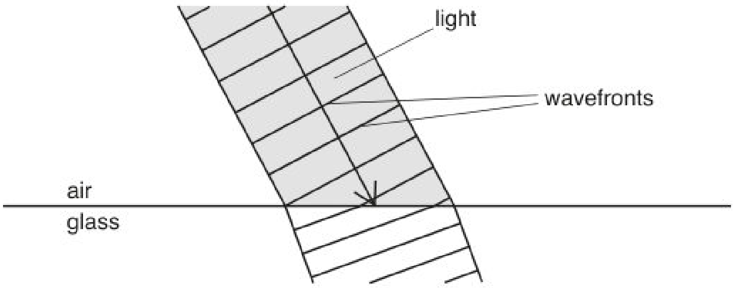


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

....

....

....

.... [3]

[Total: 15]

Mark scheme — 5054/21 Q10

Question	Answer	Marks
9(b)(iii)	it contracts **or** its density increases	B1
	it sinks	B1
	less dense / warmer water rises **or** sets up a convection current	B1
9(c)	14(°C) **or** 0.25 (kg) **or** 250 / 1000 (kg) seen	C1
	(Q =) $mc\Delta t$ **or** $0.25 \times 4200 \times (21 - 7)$ **or** $0.25 \times 4200 \times 14$	C1
	14 700 J **or** 15 000 J	A1
9(d)(i)	any **two** from: molecules / they move in clusters	B1
	slide over each other molecules / they move throughout the liquid	B1
9(d)(ii)	average speed / <u>kinetic</u> energy decreases	B1

Question	Answer	Marks
10(a)	3.0×10^8 m / s	B1
10(b)	red orange yellow green blue indigo violet (any order)	B1
10(c)(i)	(–)6.(0) cm	B1

Question	Answer	Marks
10(c)(ii)	any **two** rays drawn from: paraxial ray that refracts and seems to come from F_1 ray through the optical centre of lens ray that aims for F_2 but refracts and emerges paraxially	B2
	rays traced back to point	B1
	(point) labelled I **and** rest of image drawn down to the principal axis	B1
10(c)(iii)	1.7–1.9 cm	B1
10(c)(iv)	candidate's **10(c)(iii)** / 3.0 evaluated	B1
10(c)(v)	any **two** from: a real image can be projected on to a screen light actually passes through a real image on same side (of lens) as object **or** on opposite side of <u>mirror</u> to object	B2
10(c)(vi)	correction of short-sight / myopia	B1
10(d)	light travels more slowly in glass **or** light changes speed	B1
	one side / left-hand side of wavefront slows down first	B1
	wavelength decreases **or** wavefront travels a shorter distance in the same time	B1

Oct/Nov 5054/22

Question 5 · included (a), (b)(i)

not cleaning

(a) Describe what is meant by *longitudinal*.

....

....

.... [2]

(i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = ... [2]

Mark scheme — 5054/22 Q5

Question	Answer	Marks
3(a)(i)	(nuclear) fusion (reactions) **or** nuclei fuse	B1
	small nuclei / hydrogen nuclei joining together (and release energy) **or** larger nucleus produced	B1
3(a)(ii)	infra-red (radiation) **or** ultraviolet (radiation) **or** (visible) light	B1
3(b)	black surfaces are (good) absorbers	M1
	of (infra-red) radiation / heat / (visible) light / thermal energy	A1

Question	Answer	Marks
4(a)	any **two** from: only occurs at the surface / no bubbles produced occurs at any temperature produces cooling	B2
4(b)	molecules escape (from liquid)	B1
	fast(est) molecules / molecules with great(est kinetic) energy escape **or** slow(est) molecules / molecules with small(est kinetic) energy remain **or** (average) speed of remaining molecules decreases **or** molecules gain (kinetic) energy and escape	B1
4(c)(i)	rate of evaporation increases (with increasing area)	B1
4(c)(ii)	rate of evaporation decreases with decreasing temperature **or** takes too much time	B1

Question	Answer	Marks
5(a)	vibration / oscillation of particles **or** energy transfer **or** compressions and rarefactions	B1
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	B1

Question	Answer	Marks
5(b)(i)	(λ =) v / f **or** 1500 / 42 000	**C1**
	0.036 m **or** 3.6 cm	**A1**
5(b)(ii)	(ultrasound) vibrates (the cleaning fluid / jewellery)	**B1**
	dirt shaken off	**B1**

Question	Answer	Marks
6(a)(i)	4.0 V **c.a.o.**	**B1**
6(a)(ii)	(V =) 12 – 4.0 **or** 8.0 **or** 4.0 / 700 **or** 5.7×10^{-3}	**C1**
	(R =) V / I **or** 8.0 × 1 / (4.0 / 700) **or** 700 × (8.0 / 4.0) **or** 12 / (4.0 / 700) **or** 2100 (Ω)	**C1**
	1400 Ω	**A1**
6(b)	resistance of LDR decreases	**B1**
	current increases **or** p.d. across 700 Ω resistor / oscilloscope increases **or** p.d. across LDR decreases	**M1**
	trace moves up the screen	**A1**

Question	Answer	Marks
7(a)(i)	current **and** magnetic field / magnetic flux (experiences a force) **or** the two magnetic fields interact	**B1**
7(a)(ii)	three separate approaches **I** / **II** / **III**	
7(a)(ii)**I**	B to A	**B1**
	left-hand rule mentioned / described	**B1**
	or	

May/June 5054/21

Question 10 · included (a)

not speed-of-sound exp

(a) **(i)** Determine the wavelength of the wave in the ripple tank.

wavelength = ... [2]

Mark scheme — 5054/21 Q10

Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	**C1**
	6.4–7.2 cm	**A1**
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	**C1**
	number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time **or** number of oscillations of a source/particle in one second / unit time	**A1**
10(a)(iii)	2.5 Hz	**B1**
10(a)(iv)	moved up and down	**B1**
	regularly **or** at constant frequency / 2.5 times a second	**B1**
10(a)(v)	*speed* smaller / decreases **and** *wavelength* smaller / decreases	**B1**
10(b)(i)	tape measure **or** trundle wheel **or** microphones	**B1**
	stopwatch **or** timer	**B1**
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	**B1**
	stop timer on hearing sound	**B1**
	measure (large) distance between students	**B1**
10(b)(iii)	both between and including 1000 and 10 000 m/s	**B1**
	solid speed > liquid speed	**B1**

2018

Oct/Nov 5054/21

Question 9 · included (a), (b)

not TIR

(a) State how a longitudinal wave differs from a transverse wave.

....

.... [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

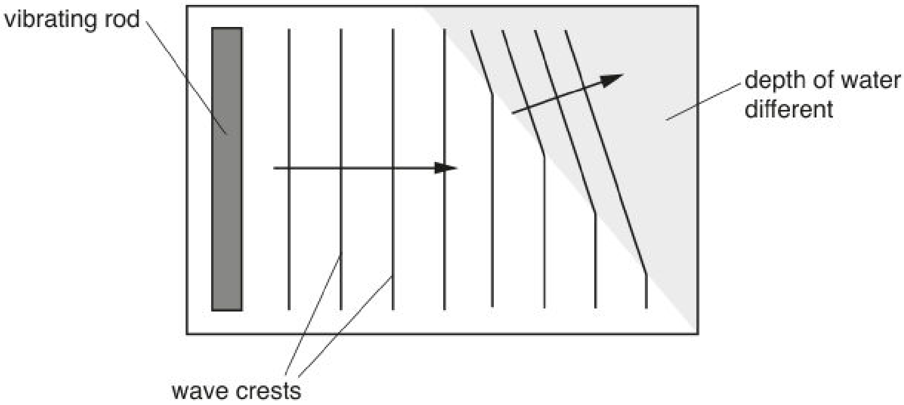


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

Mark scheme — 5054/21 Q9

5054/21 Cambridge O Level – Mark Scheme **PUBLISHED** October/November 2018

Question	Answer	Marks
8(d)(ii)	from chemical (energy) **or** chemical (energy) as first term	**B1**
	to gravitational potential (energy) **or** to thermal / heat (energy) as last term	**B1**
	to thermal / internal (energy) / heat **and** thermal / internal (energy) / heat	**B1**
8(e)	no resultant force **or** forces balance / cancel **or** tension equals weight (of bucket) **or** upwards force equals downward force	**B1**

Question	Answer	Marks
9(a)	the vibration direction is parallel to the wave / energy travel direction **or** longitudinal waves have compressions and rarefactions **or** transverse waves have crests and troughs **or** longitudinal waves cannot be polarised	**B1**
9(b)(i)	(f =) v / λ **or** 0.17 / 0.019	**C1**
	8.9 Hz	**A1**
9(b)(ii)	(frequency) stays constant **or** does not change	**B1**
9(b)(iii)	(speed) decreases	**B1**
	(Fig. 9.1 shows that) wavelength decreases **or** refraction is towards normal **or** $i > r$ **or** top of wave (in shaded region) lags behind bottom of wave	**B1**
9(c)(i)	electromagnetic **and** transverse underlined	**B1**
9(c)(ii)	$n = \sin(i) / \sin(r)$ **or** (r =) $\sin^{-1}(\sin(i) / n)$ **or** (r =) $\sin^{-1}(\sin(60°) / 1.6)$	**C1**
	33(°)	**A1**

 Page 8 of 10

Question	Answer	Marks
9(c)(iii)	1 $n = 1 / \sin(c)$ **or** (c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$	C1
	39°	A1
	2 total internal reflection	B1
	$\theta > c$ **or** 57(°) > 39(°) **or** angle of incidence greater than the critical angle	B1
	3 reflected ray at P (correct angle by eye)	B1
	light in air at 30° (by eye) to vertical surface	B1

Question	Answer	Marks
10(a)(i)	same number of protons **or** same number of electrons	B1
10(a)(ii)	different number of neutrons	B1
10(b)(i)	${}^{4}_{2}\alpha$	B1
	${}^{\cdots}_{92}U$	B1
	${}^{235}_{\cdots}U$	B1
10(b)(ii)	(no. of half-lives =) $1.2 \times 10^5 / 2.4 \times 10^4$ **or** 5 (half-lives)	C1
	relevant halving **or** $1 / 2^5$ **or** 1 / 32 **or** 6400 / 32	C1
	200 counts / second	A1

Oct/Nov 5054/22

Question 9 · included (b), (c)

not sound vs ultra

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

.... [1]

Mark scheme — 5054/22 Q9

Question	Answer	Marks
8(c)	any **two** from: smaller time to drop to zero velocity / hit ground line not straight **or** velocity does not change uniformly **or** gradient not constant smaller area under (first part of) graph **or** less distance travelled slower final velocity initial downward gradient steeper	B2

Question	Answer	Marks
9(a)	frequency of sound wave small(er) **or** its frequency is less than 20 000 Hz	B1
9(b)(i)	transmission of energy	B1
	(through a medium) with no net movement of medium **or** by vibrating particles	B1
	vibrations parallel (and antiparallel) to wave / energy travel direction **or** cannot be polarised	B1
9(b)(ii)	1 two centres of rarefactions labelled *R*	B1
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	B1
	3. (v =) $f\lambda$ **or** 25 000 × 0.047 / 0.048 / 0.049 **or** 25 000 × 4.7 / 4.8 / 4.9 **or** 25 000 × 47 / 48 / 49	C1
	1200 m / s	A1
9(c)(i)	decreases	B1
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid **and** no intermediate / extra lines between the correct lines	B1
	at least four compressions in air parallel to each other	B1
	at least four straight lines at shallower angle from horizontal **and** slope correct	B1

Question	Answer	Marks
9(d)	object (to be cleaned) immersed in liquid / solvent	B1
	object / liquid agitated / vibrated by ultrasound	B1
	dirt (particles) shaken off **or** dislodges / removes dirt	B1

Question	Answer	Marks
10(a)	electrons **c.a.o.**	M1
	towards the ammeter **or** away from the negative terminal **or** towards the positive terminal	A1
10(b)(i)	thermistor **c.a.o.**	B1
10(b)(ii)	$1 / R_T = 1 / R_1 + 1 / R_2$ **or** $1 / R_T = 1 / 1.5 + 1 / 6.0$ **or** (R_T =) $R_1R_2 / (R_1 + R_2)$ **or** $1.5 \times 6.0 / (1.5 + 6.0)$	C1
	1.2 (Ω)	C1
	2.5 Ω	A1
10(b)(iii)	(I =) V / R **or** 12 / 2.5	C1
	4.8 (A)	A1
10(b)(iv)	$I_A = I_R + I_Z$	B1
10(c)	resistance of Z / thermistor decreases	B1
	resistance of parallel combination decreases **or** total resistance (of circuit) decreases **or** current increases	B1
	voltage (across 1.3 Ω) increases	B1
	trace moves towards top of screen / upwards	B1

May/June 5054/21

Question 11 · included (a), (b)

10 **(a)** **(i)** Describe two differences between boiling and evaporation.

1. ....

....

....

2. ....

....

.... [4]

(b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16 °C
temperature of water after heating = 37 °C
volume of water collected in 1.0 minute = $4.6 \times 10^{-3}\,m^3$
specific heat capacity of water = 4200 J / (kg °C)
density of water = $1000\,kg/m^3$

(i) Calculate the mass of water leaving the shower in 1.0 s.

mass = ...[2]

Mark scheme — 5054/21 Q11

Question	Answer	Marks
11(a)(i)	tank with water	**B1**
	dipper (bar or point) touches or in surface of water	**B1**
	light source **/** stroboscope / video	**B1**
	dipper moves up and down (to create wave)	**B1**
11(a)(ii)	observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object	**B1**
	moves up and down	**B1**
11(a)(iii)	line joining points	**M1**
	that are at the same point in the motion	**A1**
11(b)(i)	*difference*: Q has a greater / different amplitude or phase	**B1**
	similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down	**B1**
11(b)(ii)	time 300 (ms) or 0.3 (s)	**C1**
	(f=) 1 / t numerical or algebraic in any form	**C1**
	3.3 Hz	**A1**
11(b)(ii)	2 (λ =) v / f **or** v = dt – numerical or algebraic in any form	**C1**
	0.060 m	**A1**

May/June 5054/22

Question 5 · included (b)

only long/trans + compression

(b) The object has a constant speed for some of the time.

Mark scheme — 5054/22 Q5

Question	Answer	Marks
5(a)	(d=) s × t in any form algebraic or numerical,	C1
	3.6 × 10^7 m	A1
5(b)(i)	1st two columns correct (sound longitudinal and microwaves transverse)	B1
	3rd column correct microwaves electromagnetic	B1
5(b)(ii)	Layers / molecules / particles / spring coils close together **or** high(er) pressure (than atmospheric)	B1

Question	Answer	Marks
6(a)	**any** two of • ray through middle of lens undeviated • ray parallel to axis passes through focus, 3 cm from lens • ray through focus on left of lens parallel to axis after lens	B2
6(b)	inverted **or** real	B1
6(c)	image size / object size **or** image distance / object distance numerical or algebraic	C1
	rays from lens intersect on diagram (for image) **and** 1.3–1.7	A1
6(d)	projector **or** photographic enlarger	B1

2017

Oct/Nov 5054/21

Question 10 · included (c)(i)

only v=fλ

(c) The speed of electromagnetic radiation in a vacuum is 3.0×10^8 m/s.

A television remote controller uses infra-red radiation of wavelength 9.4×10^{-7} m.

(i) Calculate the frequency of this radiation.

frequency = ... [3]

(ii) Explain how infra-red radiation is used in television remote controllers.

....

....

....

.... [2]

Mark scheme — 5054/21 Q10

Question	Answer	Marks
10(a)	P - gamma(-rays) **or** γ(-rays)	
	Q - ultraviolet (radiation)	
	R - microwaves	
	any one correct	**C1**
	all three correct	**A1**
10(b)	P **and** X-rays **and** Q ticked	**B1**
10(c)(i)	$(f =)c / \lambda$ **or** $3.0 \times 10^8 / 9.4 \times 10^{-7}$	**C1**
	3.2×10^N	**C1**
	3.2×10^{14} Hz	**A1**
10(c)(ii)	infra-red / radiation / signal / wave emitted by control **and** received at set	**B1**
	infra-red / radiation / signal / wave is encoded **or** is decoded	**B1**
10(d)(i)	normal indicated **and** angle of incidence indicated	**B1**
10(d)(ii)	$n = \sin i / \sin r$ **or** $1.5 = \sin 57(°) / \sin r$ **or** $(r =)\sin^{-1}(\sin 57(°) / n)$ **or** $\sin^{-1}(\sin 57(°) / 1.5)$	**C1**
	34°	**A1**
10(d)(iii)	**1** no change	**B1**
	2 **3** decreases **and** decreases	**B1**

Question	Answer	Marks
10(d)(iv)	ray in glass between normal and continuation of the incident ray	**B1**
	ray in air between continuation of the refracted ray and side of prism	**B1**

Question	Answer	Marks
11(a)(i)	any suitable solid insulator (e.g. nylon, plastic, glass, rubber, polystyrene)	**B1**
11(a)(ii)	positive charges near to rod	**B1**
	negative charges opposite rod **and** equal in number **and** 7 or fewer	**B1**
11(a)(iii)	**1** electrons / negative charges flow towards earth	**B1**
	repelled (by negative charge on rod)	**B1**
	(sphere) becomes positive	**B1**
	2 flow of electrons / negative charge **and** (in direction) earth to sphere	**B1**
11(b)(i)	$1 / R = 1 / R_1 + 1 / R_2$ **or** $R_1 R_2 / (R_1 + R_2)$ **or** $1 / R = 1 / 15 + 1 / 60$ **or** $15 \times 60 / 75$ **or** $15 \times 60 / (15 + 60)$	**C1**
	12 (Ω) **or** 0.083 (Ω)	**C1**
	30 Ω	**A1**
11(b)(ii)	$(I =)V / R$ **or** 7.5 / 30	**C1**
	0.25 A	**A1**

Oct/Nov 5054/22

Question 6 · included (a), (b), (c)

(a) State what is meant by the term *echo*.

....

....[1]

(b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

(c) The speed of sound in air is 330 m/s.

Calculate the wavelength of this sound.

wavelength = ...[2]

Mark scheme — 5054/22 Q6

Question	Answer	Marks
5(a)	joining together of (small) nuclei (to make bigger nuclei)	B1
	energy released	B1
	hydrogen (used) **or** helium (produced)	B1
5(b)	electromagnetic radiation / infra-red / light / ultraviolet	B1
	travels through vacuum **or** no medium needed	B1

Question	Answer	Marks
6(a)	reflection of sound	B1
6(b)(i)	decreases	B1
6(b)(ii)	does not change	B1
6(c)	(λ =) c / f **or** 330 / 3700	C1
	0.089 m **or** 8.9 cm **or** 89 mm	A1

May/June 5054/21

Question 10 · included (b), (c)(i), (c)(iii)

not mirror; not hearing

(b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

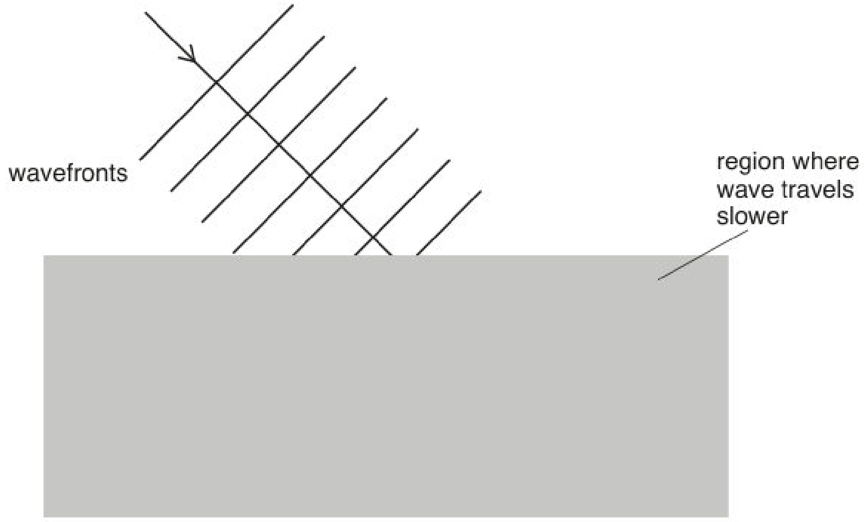


Fig. 10.2

(i) State what is meant by *wavefront*.

....

....[2]

(i) Calculate the speed of sound in air.

speed = ...[2]

(ii) 1. State the range of frequencies that can be heard by a healthy human ear.

....[1]

2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = ...[1]

(iii) Describe a simple experiment to show that sound waves obey the law of reflection. You may draw a diagram if you wish.

....

....

....

....[3]

Mark scheme — 5054/21 Q10

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	B1
	image in correct position shown by continuation of rays behind mirror	B1
10(a)(ii)	virtual **or** upright **or** laterally inverted **or** same size (as object)	B1
10(b)(i)	line (joining points)	C1
	line joining points / particles on wave with same phase **or** line joining points along a crest etc.	A1
10(b)(ii)1	correct angle to surface ± 7° with correct orientation **and** similar wavelength (by eye)	B1
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	B1
	showing smaller and constant wavelength with wavefronts deviated in correct direction	B1
10(c)(i)	(v =) fλ **or** 2(000) × 16	C1
	32 000 cm / s **or** 320 m / s	A1
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency **and** 15–30 kHz as the highest	B1
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	B1
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	B1
	stated detector **or** stated reflector	B1
	detector moved to find maximum loudness **or** angles *i* and *r* measured with suitable experiment	B1

May/June 5054/22

Question 6 · included (a), (b)(i)

not hearing

6 **(a)** Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

....

....

....

....

....[2]

(i) Calculate the wavelength of the sound.

wavelength = ...[2]

Mark scheme — 5054/22 Q6

5054/22 Cambridge O Level – Mark Scheme **PUBLISHED** May/June 2017

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro **and** in direction of wave **or** has compressions and rarefactions	**B1**
	transverse – vibration / oscillation / movement up and down **and** at right angles to wave **or** has crests and troughs	**B1**
6(b)(i)	(λ =) v / f or 330 / 3800	**B1**
	0.087 m **or** 8.7 cm	**B1**
6(b)(ii)	not heard **and** because below the range of audible frequencies **or** audible range is 20 – 20 000 Hz **or** too low a pitch / frequency	**B1**

Question	Answer	Marks
7(a)(i)	current in coil (at right angles) in a magnetic field (of magnet) **or** left-hand rule mentioned	**B1**
7(a)(ii)	reverses / changes direction of current (in coil)	**B1**
	reverses current every half turn / when coil is vertical **or** reverses forces (on side AB / CD) **or** keeps forces in same direction for wire on one side	**B1**
7(b)(i)	(E =)VIt or 2 × 12 × 8 **or** E = Pt **and** P = VI **or** E = VQ **and** Q = It	**C1**
	190 **or** 192 J	**A1**
7(b)(ii)	73% **or** 0.73	**B1**

 Page 5 of 10

2016

Oct/Nov 5054/21

Question 5 · included (a), (b)

not (c) production/vacuum

(a) State what is meant by the *frequency* of a sound.

....

....

....[1]

(b) **(i)** The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength = ...[2]

Mark scheme — 5054/21 Q5

5	(a)		number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time	B1	
	(b)	(i)	($\lambda =$) c / f **or** 330 / 2200 0.15 m	C1 A1	
		(ii)	**1** no change **and** **2** increases	B1	
	(c)	(i)	**1** loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules) **2** compressions **and** rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on	B1 B1 B1	
		(ii)	fewer / no molecules / particles **and** less / no energy / vibration transferred	B1	[8]
6	(a)	(i)	X N-pole Y S-pole **and** Z N-pole	B1 B1	
		(ii)	they touch / move towards each other **and** opposite poles attract	B1	
	(b)		any sensible use: starting-motor circuit; with a logic gate; nuclear power station corresponding explanation: current too large for dash-board switch; current too small to power device; too dangerous to reach switch	B1 B1	[5]
7	(a)	(i)	supplies the (mains) e.m.f. / voltage	B1	
		(ii)	to complete the circuit / is at 0 V	B1	
	(b)	(i)	the circuit / supply is cut / broken **or** current stops fuse melts / blows / burns	B1 B1	
		(ii)	live wire when it cuts the circuit / melts no part of the appliance is live / no shock	B1 B1	[6]
					[45]

Oct/Nov 5054/22

Question 10 · included (a)

only $v=f\lambda$

(a) Calculate the speed of the skier after 5.0 s.

speed = ...[2]

Mark scheme — 5054/22 Q10

	(ii)	1	cylinder is magnetised (by induction) top (of cylinder) is an S-pole unlike poles attract **or** S-pole attracts N-pole	B1 B1 B1	
		2	it does not (remain in contact**)** **and** iron is temporary / soft magnetic material / core (and cylinder) lose magnetisation	B1	[4]
					[15]
10	**(a)**	**(i)**	3.0×10^8 m / s	B1	
		(ii)	(λ =)c /f **or** $3.0 \times 10^8 / 4.3 \times 10^{14}$ 7.0×10^{-7} m	C1 A1	[3]
	(b)	**(i)**	decreases	B1	
		(ii)	sin(i) = n × sin(r) **or** 1.5 × sin(30°) **or** 0.75 49°	C1 A1	
		(iii)	41°	B1	[4]
	(c)	**(i)**	dispersion at **both** surfaces **and** refractions in correct direction violet / blue light below the red light shown	B1 B1	
		(ii)	spectrum **or** band of (continuous) colours **or** colours of rainbow red, orange, yellow, green, blue, (indigo, violet)	B1 B1	
		(iii)	**1** X marked above red **2** it is / black surfaces are good absorbers (of IR radiation)	B1 B1	[6]

(d)	intruder / human being emits IR		IR beam broken		IR reflected	B1	
		or		**or**			
	intruder warm **or** IR detected		does not reach detector		change detected	B1	[2]
							[15]

May/June 5054/21

Question 5 · included (a), (b)

5 **(a)** Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

wave	direction of vibration	type of wave
light	perpendicular to the direction of transfer of the energy	transverse
sound		
water		

[2]

Fig. 5.1

(b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.

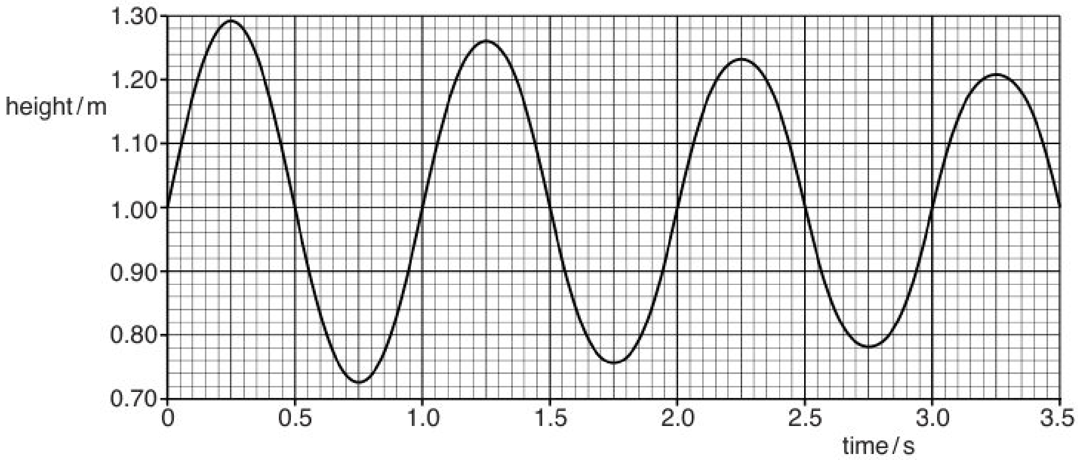


Fig. 5.2

Mark scheme — 5054/21 Q5

	(b)	**(i) 1**	difference between smallest and largest temperature **or** from 0 to 100 °C	B1
		(i) 2	small/moderate distance between (thermometer) marks **or** for a given temperature change there is a small expansion of liquid / distance (along scale) / change in thermometric property **or** cannot measure small temperature <u>difference / change</u>	B1
		(ii)	• use liquid that expands more • smaller bore / thinner tube • more mercury (in bulb) **or** use larger bulb	B1
5	**(a)**		*sound:* along or parallel (to transfer of energy or wave) **and** longitudinal	B1
			water: perpendicular **and** transverse	B1
	(b)	**(i)**	0.29 – 0.28 m	B1
		(ii)	<u>time / period</u> for one wave(length) / cycle constant **or** each oscillation / cycle takes one second	B1
6	**(a)**		angle of incidence	B1
			smallest angle for light to be totally internally reflected **or** largest angle (of incidence) for ray to be refracted / emerge **or** when light emerges along surface **or** when angle of refraction is 90°	B1
	(b)	**(i)**	n = 1 / sinC algebraic or numerical 2.5 **or** 2.46 **or** 2.458(59)	C1 A1
		(ii)	*left hand diagram* ray refracts away from normal and emerges into air at bottom left surface	B1
			right hand diagram reflected horizontal ray (by eye)	B1
			right hand diagram rest of ray completely correct to emerge into air at top face without refraction (by eye)	B1
7	**(a)**		(current in coil) creates magnetic field **or** current is at right angles to magnetic field (of permanent / cylindrical magnets)	C1
	(b)		into and out of magnet **or** left and right **or** backwards and forwards	B1
			current is one way then reverses (so reverses force)	B1

Question 7 · included (c)

only v=fλ

(c) The loudspeaker produces sound of frequency 500 Hz. The speed of sound is 320 m/s. Calculate the wavelength of the sound produced.

wavelength = ...[2]

 5054/21/M/J/16

Mark scheme — 5054/21 Q7

	(b)	**(i) 1**	difference between smallest and largest temperature **or** from 0 to 100 °C	B1
		(i) 2	small/moderate distance between (thermometer) marks **or** for a given temperature change there is a small expansion of liquid / distance (along scale) / change in thermometric property **or** cannot measure small temperature <u>difference / change</u>	B1
		(ii)	• use liquid that expands more • smaller bore / thinner tube • more mercury (in bulb) **or** use larger bulb	B1
5	**(a)**		*sound:* along or parallel (to transfer of energy or wave) **and** longitudinal	B1
			water: perpendicular **and** transverse	B1
	(b)	**(i)**	0.29 – 0.28 m	B1
		(ii)	<u>time / period</u> for one wave(length) / cycle constant **or** each oscillation / cycle takes one second	B1
6	**(a)**		angle of incidence	B1
			smallest angle for light to be totally internally reflected **or** largest angle (of incidence) for ray to be refracted / emerge **or** when light emerges along surface **or** when angle of refraction is 90°	B1
	(b)	**(i)**	n = 1 / sinC algebraic or numerical 2.5 **or** 2.46 **or** 2.458(59)	C1 A1
		(ii)	*left hand diagram* ray refracts away from normal and emerges into air at bottom left surface	B1
			right hand diagram reflected horizontal ray (by eye)	B1
			right hand diagram rest of ray completely correct to emerge into air at top face without refraction (by eye)	B1
7	**(a)**		(current in coil) creates magnetic field **or** current is at right angles to magnetic field (of permanent / cylindrical magnets)	C1
	(b)		into and out of magnet **or** left and right **or** backwards and forwards	B1
			current is one way then reverses (so reverses force)	B1

May/June 5054/22

Question 10 · included (b)

only CRO A and T

(b) A microphone is connected to a c.r.o. to display a sound wave.

Fig. 10.2 shows the trace on the c.r.o.

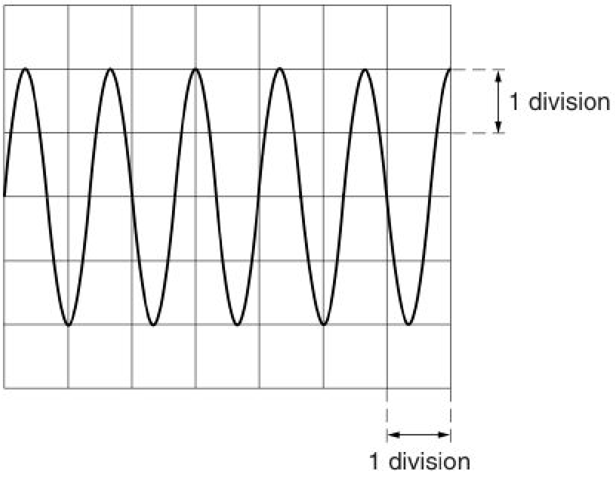


Fig. 10.2

The settings on the c.r.o. are: Y-gain 0.5 V / division; time base 2.0 ms / division.

Mark scheme — 5054/22 Q10

	(c)		current is (directly) proportional to voltage **or** voltage / current is a constant law holds for constant physical conditions / constant temperature / constant pressure / for metals	B1 B1
	(d)	**(i)**	(directly) proportional **or** (R) ∝ 1	B1
		(ii)	inversely proportional **or** (R) ∝ 1 / A	B1
	(e)		1st band orange 2nd and 3rd bands both black	B1 B1
10	**(a)**	**(i)**	B – anode D – filament **or** heater E and F–Y plates or X plates in either order	B1 B1 B1
		(ii)	**1** attract electrons **or** gives electrons speed / K.E.	B1
			2 heats up cathode **or** gives electrons energy to escape (metal / cathode) **or** causes / allows thermionic emission	B1
		(iii)	kinetic energy to light **or** electrical energy to light	B1
		(iv)	voltage/charge is applied to the X-plates / vertical plates **or** turn on time base (steadily) increasing voltage / charge applied to plate(s) **or** saw tooth voltage applied **or** electrons attracted / repelled by plate(s) or by the electric field between them	B1 B1
	(b)	**(i)**	**1** 1(.0) V	B1
			2 one wave 1.3–1.4 squares **or** 3 waves in 4 squares 2.6–2.8 ms	C1 A1
			3 (f =) 1 / T numerical or algebraic 345–400 Hz	C1 A1
		(ii)	smaller amplitude shown larger period shown	B1 B1

2015

Oct/Nov 5054/21

Question 7 · included (a), (b)

not (c) ultrasound use

(a) Describe one difference between a compression and a rarefaction.

....

....

.... [1]

(b) In a particular medium, the speed of both audible sound and ultrasound waves is *v*.

Mark scheme — 5054/21 Q7

5	(a)		**temperature** at which a liquid becomes a gas	B1	
	(b)	(i)	molecules close together / touching **or** closer than in gas randomly arranged **or** irregular structure	B1 B1	
		(ii)	to separate / increase the distance between molecules work done against (intermolecular) forces **or** supply p.e. **or** break bonds	B1 B1	[5]
6	(a)		**distance** from (optical) centre to focal point (principal focus)	B1	
	(b)	(i)	**both** Fs correctly positioned at ±1 mm	B1	
		(ii)	**two** of: paraxial ray to lens through focal point to image ray through optical centre ray through focal point and then paraxial to image (**ign.** arrows) X at crossing point of rays	 M2 A1	
		(iii)	3.4–3.8 cm	B1	[6]
7	(a)		at compression: molecules closer together **or** pressure higher **or** vice versa for rarefaction	 B1	
	(b)	(i)	$v = f\lambda$ **or** in words	B1	
		(ii)	larger **and** because the frequency is lower	B1	
	(c)		states one use (e.g. prenatal scanning) basic idea (e.g. ultrasound reflects off foetus)	B1 B1	[5]
8	(a)	(i)	$(I =)V/R$ **or** 12 / (6000 + 2000) **or** 12 / 8000 **or** 12 / 2000 **or** 12 / 6000 **or** in **(ii)** $(V =)IR$ **or** 0.0015 × 6000 1.5 mA	 C1 A1	
		(ii)	9.0 V	B1	
	(b)		(reading / it) increases resistance of LDR falls	B1 B1	
	(c)		light meter / sensor **or** automatic light switch **or** something sensible	B1	[6]
					[45]

May/June 5054/21

Question 4 · included (a), (b), (c)

(a) Describe the movement of the particles of the string.

....

....

....

....[2]

(b) Determine the wavelength of the wave.

wavelength = ...[1]

(c) The speed of the wave is 2.0 m/s.

Mark scheme — 5054/21 Q4

1	(a)	(i)	60 m	B1
		(ii)	12 s	B1
	(b)	(i)	straight line from origin to 200 m at 40 s any line straight or curved from (40,200) to (60,500)	B1 B1
		(ii)	$s = d / t$ or 500 / 60 8.3 m / s	C1 A1
2	(a)	(i)	force moves through a distance (in same direction)	B1
		(ii)	chemical (potential) energy	B1
	(b)	(i)	480 Nm	B1
		(ii)	attempt to apply moments with two forces and distances 400 N	C1 A1
3	(a)		Pa **or** N / m^2 **or** cm of mercury **or** atmosphere(s)	B1
	(b)		correct points plotted at ($0.5V_0$, $2P_0$) and ($2V_0$, $0.5P_0$) curve through points of decreasing gradient	B1 B1
	(c)		molecules hit sides / piston	B1
			more molecules hit per second / hit more frequently molecular impacts create large(r) **force** (upwards on piston)	B1 B1
4	(a)		oscillate / vibrate stated or described transverse movement described	B1 B1
	(b)		0.40 m	B1
	(c)	(i)	$v = f\lambda$ or (f =) v / λ or 2 / (b) 5.0 Hz	C1 A1
		(ii)	clear attempt to draw wave moved along 0.20 m to right	B1
5	(a)		sin i / sin r or sin 50 / sin 30 1.5(321)	C1 A1

May/June 5054/22

Question 9 · included (a), (b), (c), (d)

(a) Explain what is meant by the *frequency* of a wave.

....

....

.... [2]

(b) **(i)** Determine the wavelength of the wave in deep water.

.... [1]

(c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

(i) State and explain how this affects the wavelength of the wave.

....

....

.... [2]

(d) Sound is also a wave.

Mark scheme — 5054/22 Q9

7	**(a)**	to provide a complete circuit (with live) **or** to pass current back to mains **or** provide a return path for the current	B1
	(b)	current / charge / electrons flow to earth / earth wire / ground (when live touches case) fuse melts / blows **and** disconnects circuit / cuts live / stops current	B1 B1
	(c)	doubly insulated **or case / body** made of plastic / insulator / not made of metal **or** user cannot touch metal	B1
	(d)	(circuit breaker) • turns off / acts fast(er) • can be reset • easy to see it has tripped / switched • can detect small difference between live and neutral currents / small (leakage) current to earth	B1
8	**(a)**	left column both 1 right column 0 **and** 1	B1 B1
	(b)	(at least one of the atoms) contain same number of electrons and protons **or have** 1 electron and 1 proton charge on electron and proton opposite **or** electron negative **and** proton positive **or** charge on electron neutralises/cancels/balances proton charge neutrons have no charge	B1 B1 B1
9	**(a)**	number of waves (that pass a point) **or** number of oscillations (passing a point) in unit time **or** per second **or** in 1 second	M1 A1
	(b) (i)	1.5 cm	B1
	(ii)	$(v =)f\lambda$ **or** 5×1.5 seen **or** $(s=)d/t$ **and** $f = 1/t$ 7.5 cm / s	C1 A1
	(c) (i)	wavelength decreases travels a shorter distance in the same time **or** frequency stays the same (and $v = f\lambda$)	B1 B1